METRIC

MIL-C-71164 (AR)
11 March 1994

MILITARY SPECIFICATION

CARTRIDGE, 25MM, DUMMY, M794

PERFORMANCE SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the 25mm M794 Dummy Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1305

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SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition, General Specification for
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-71136	-	Process Specification, Ammunition, 25mm, Packaging, Packing and Marking

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance for
MIL-STD-644	-	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-651	-	Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT

12013880	-	Cartridge, 25mm, Dummy, M794

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INSPECTION

9391400	- Interface Control Drawing, Max Profile and Alignment Cartridge, M792, M793 and HPT
12013695	- Characteristics, Linked Belt 25mm, M28

PACKAGING

5167214	- Shipping and Storage Container, 25mm Ammo, Belted with M28 Links Assembly
12013668	- Container, Shipping and Storage, Plastic, Linked, 25mm Ammunition
12013671	- Packing and Marking 50 Cartridges, Linked Shipping and Storage Container Cartridge, 25mm
12013674	- Packing and Marking 80 Cartridges, Unlinked, Shipping and Storage
12013698	- Protective Cover - 25mm Round
12929427	- Packing and Marking for Container, Ammunition, PA125, for Linked 25mm Ammunition

PUBLICATIONS

AS12013566	- 25 Millimeter Ammunition Ballistic Test Methods

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.2 Cartridge. The cartridge and all components thereof shall comply with all requirements specified on Drawing (Dwg) 9391400, all associated drawings, and with all requirements specified in applicable specifications and drawings.

3.3 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.4 Structural integrity. The cartridge shall withstand 50 non-firing cycles through a temperature range of -32° to 52°C after 4 hours of stabilized conditioning temperature. The weapon shall be cycled at a rate of 200 ± 25 shots per minute (spm) in both AP and HE dual feeders without gun stoppage. The non-firing cycle includes feeding, chambering and ejection.

3.5 Salt spray (fog). The exterior surface of the cartridge shall show no evidence of corrosion (red rust) or swelling after exposure to a 5 percent solution of atomized salt fog for 48 hours when tested as specified in paragraph 4.5.5.

3.6 Temperature-humidity. The cartridge shall be capable of withstanding a humidity environment in accordance with paragraph 4.5.6.

3.7 Projectile torque. The projectile shall withstand the minimum torque 6.78 newton-meter (60 in-lbs).

3.8 Projectile extraction. The tensile force necessary to extract the projectile from the cartridge case shall exceed 452 newtons.

3.9 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, scratches, dents, folds, wrinkles, buckles, dirt, grease, oil, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and clearly defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for

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the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-2550 apply and form a part of this specification. Terms, definitions and standard quality assurance provisions included in MIL-A-48078 are applicable in their entirety if not in conflict with MIL-A-2550 or unless otherwise specified herein or in the contract. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Cartridge, 25mm, Dummy, M794	12013880	350

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4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71164 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Cartridge, 25mm, Dummy, M794		12013880
			NEXT HIGHER ASSEMBLY
			N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Examination for defects	350 1/	3.2	4.4.2.1, except Majors 105, 106, 107, 108, 109
	Salt spray (fog)	25	3.5	4.5.5
	Temperature-Humidity	25	3.6	4.5.6
	Projectile Torque	50	3.7	4.5.2
	Projectile Extraction	50 2/	3.8	4.5.3
	Gun Cycle			
	-32 deg + 3 deg C Tests (50 cycles each feeder)	100	3.4	4.5.4 3/
	52 deg - 3 deg C Tests (50 cycles each feeder)	100	3.4	4.5.4 3/

NOTES:

1/ After this test, the 350 cartridges are to be divided for subsequent tests as follows: 25 Salt Spray, 25 Temperature Humidity, 50 Projectile Torque, 200 Gun Cycle Drop Tests, 50 Projectile Extraction. 2/ Single sample size may be used and test combined provided no movement is experienced in the torque test, which shall be performed prior to extraction testing. 3/ The test shall consist of 2 belts of 50 linked cartridges each. One belt shall start from the left side for 50 cycles, followed by 50 cycles on the right side without temperature reconditioning. The other belt shall follow a similar sequence, starting from the right side.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and consist of the following:

a. Cartridges manufactured by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and the same specification and specification revisions.

b. Packaging materials as required by paragraph 5.2.

c. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sample inspection.

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71164 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Cartridge, 25mm, Dummy, M794		12013880
			NEXT HIGHER ASSEMBLY
			N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101.	Weight	Level III	3.2	Balance 4/
102.	Overall length, max	Level III	3.2	Gage
103.	Profile and alignment	Level III	3.2	Gage/9391400 1/
104.	Looseness between projectile and cartridge case		3.2	Visual/Manual 3/
105.	Projectile Torque	20 Acc-0 Rej-1	3.7	4.5.2
106.	Projectile Extraction	20 Acc-0 Rej-1	3.8	4.5.3 5/
107.	Gun cycle/Drop test	100 Acc-0 Rej-1	3.4	4.5.4 2/
108.	Salt spray (fog)	20 Acc-0 Rej-1	3.5	4.5.5
109.	Temperature and Humidity	Level III	3.6	4.5.6

NOTES:

1/ A dead weight load of 40 lb max may be used to insert the cartridge into the gage. 2/ The test shall consist of 2 belts of 50 linked cartridges each. One belt shall start the test from the left side and the other belt shall start from the right side. 3/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects, as applicable. 4/ The weight of the dummy cartridge shall be 498 ± 25 grams. 5/ Single sample size may be used and tests combined provided no movement is experienced in the torque test, which shall be performed prior to the extraction testing.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71164 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 25mm, Dummy, M794	2 OF 2	12013880
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201.	Marking incorrect, incomplete, illegible, or missing	Level IV	3.2	Visual 3/
202.	Workmanship	Level IV	3.9	Visual 3/

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71164 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge, 25mm, Dummy, M794, in 50 Round Linked Belt, 25mm, M28 for use in Container	1 OF 1	12013695/12013671
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101.	Height from base of cartridge to forward tab, max.	Level III		Gage/12013695
102.	Dimple not engaged in cartridge case groove, two places	100 percent		Visual 1/ 12013695
103.	Improper fit of protective cover	Level III	12013698	Visual/Manual 1/
Minor				
201.	Evidence of poor workmanship	Level IV	3.9	Visual 2/

NOTES:

1/ MIL-STD-644 shall apply in defining and evaluating visual defects, as applicable.

2/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects, as applicable.

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4.4.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components and assemblies have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.2 Projectile torque. Each cartridge of the test sample shall be marked with a line extending longitudinally across the projectile/case joint. The required torque shall be applied slowly to the projectile body in a counterclockwise direction. Any cartridge that has movement of the projectile body with respect to the cartridge case, detectable by misalignment of the line, shall be classed defective. Cartridges subjected to this test shall not be returned to the lot.

4.5.3 Projectile extraction. The method of test shall be as specified in AS12013566, Section 12. Failure to satisfy any requirement shall be cause for rejection of the lot.

4.5.4 Gun cycle/drop test. The sample cartridges shall be assembled into a linked belt and temperature stabilized a minimum of 4 hours at the specified temperatures before dry-cycling at 200 ± 25 cycles per minute rate through the M242 weapon. If any stoppage is caused by a defective test weapon, as determined by the Government, the defect shall be recorded for informational purposes. If such defect is not caused by a defective weapon, the cartridge shall be classed defective. All cartridges subjected to this test shall not be returned to the lot.

4.5.5 Salt spray test. The sample cartridges shall be exposed to a 5 percent solution of atomized salt fog for a period of 48 hours. The test shall be conducted in accordance with test method 107 of MIL-STD-331. Cartridges subjected to this test shall not be returned to the lot.

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4.5.6 Temperature-humidity. The test and inspection of the sample cartridges shall be performed in accordance with MIL-STD-810, Method 507, Procedure III.

5. PACKAGING

5.1 Linked belt or protective cover. The cartridge shall be linked into a belt in accordance with Dwg. 12013695 or have a protective cover per Dwg. 12013698 attached, as applicable.

5.2 Packing, level A. The type of ammunition pack shall be in accordance with instructions from the procuring agency. Linked ammunition shall be packed in accordance with Dwg. 12929427 and MIL-P-71136; or Dwg. 12013671 or Dwg. 12013668 or 5167214. Unlinked ammunition shall be packed in accordance with Dwg. 12013674.

5.2.1 Packing level B. Same as 5.2.

5.2.2 Packing level C. Same as 5.2.

5.3 Marking. Marking on the packed container shall be in accordance with Dwg. 12013674 or Dwg. 12013671 or Dwg. 12013674 as applicable, or Dwg. 12013668 or 5167214.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These cartridges are intended for use in cycling the M242.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Deformed link. A deformed link is defined as a link whose shape or dimensions or both have been mutilated or altered by processing to the extent that it is unsuitable for its intended use.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Subject term (key word) listing.

Practice Loading
Practice Cycling
Practice Unloading

6.7 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AE49)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71164 (AR)	940311

3. DOCUMENT TITLE: CARTRIDGE, 25MM, DUMMY, M794

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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